

Meet Udeshi

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Ph.D. Candidate, NYU Tandon School of Engineering

Expected graduation: Summer 2027

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RESEARCH INTERESTS

My research interests span **cyber-physical and embedded systems security**, with an emphasis on scalable techniques for analyzing, validating, and securing complex software-driven systems. I develop methods for **binary analysis** and semantic reverse engineering using **symbolic execution** and **neural decompilation**, and explore **malware sandboxing** and **digital twins** for realistic, system-aware security evaluation. I also investigate **LLM-based agents** for automating security analysis and emerging threats from **AI-powered malware**.

EDUCATION

NYU Tandon School of Engineering

Ph.D., Electrical Engineering

Sep 2022–present

Control/Robotics Research Lab, Center for Cybersecurity. GPA: 3.958/4

Advisors: Prof. Farshad Khorrami and Prof. Ramesh Karri

Indian Institute of Technology Bombay (IIT Bombay)

B.Tech. and M.Tech. (Dual Degree), Electrical Engineering

July 2014–June 2019

Computer Architecture and Dependable Systems Lab. CGPA: 8.18/10

Thesis Advisor: Prof. Virendra Singh

WORK EXPERIENCE

Siemens Healthineers USA, Princeton, NJ

May 2026–Aug 2026

AI Cyber Intern, AI Factory

Contributed to ARPA-H UPGRADE to make cybersecurity patching more efficient for clinical and hospital setups. Modeled the cyber-clinical impact of unpatched vulnerabilities in critical infrastructure.

Qualcomm R&D, Bangalore, India

July 2019–July 2022

Senior Engineer, ML Compiler Team for Cloud AI100 Accelerator

Worked on key aspects of the AI100 compiler: multi-core, multi-thread data tiling, memory management, graph scheduling, and operator fusion. Developed graph optimization techniques for 2D and 3D computer vision, recommendation systems, and autonomous driving models. Contributed to the open-source *PyTorch Glow* compiler framework. Deployed a power-efficient object tracking pipeline using Kernelized Correlation Filters (KCF) on AI100.

Amazon India

Jul 2017–Aug 2017

Software Development Intern, Transportation Financial Systems

Implemented processing and sorting of 1 million+ receipts daily using DynamoDB and SQS. Automated setup of 30+ AWS resources with CloudFormation.

Google Summer of Code, Kivy

May 2016–Aug 2016

Python Native UI Framework Game Engine

Created a Python/Cython module for Tiled maps integration with the KivEnt game engine. Implemented a Cython-optimized animation system using an entity-component architecture.

JOURNAL PUBLICATIONS

Equal contributions are marked with †.

- [J1] **M. Udeshi**, P. Krishnamurthy, R. Karri, and F. Khorrami. “REMEMD: Neural Decompilation for Reverse Engineering Math Equations from Binary Executables.” *ACM Transactions on Intelligent Systems and Technology*, 17(3), Article 64, 2025.

	[J2]	M. Udeshi , P. Krishnamurthy, R. Karri, and F. Khorrami. "Tamper-Proof Network Traffic Measurements on a NIC for Intrusion Detection." <i>IEEE Transactions on Network and Service Management</i> , 22(2), 2025.	
	[J3]	M. Udeshi , P. Krishnamurthy, H. Pearce, R. Karri, and F. Khorrami. "REMaQE: Reverse Engineering Math Equations from Executables." <i>ACM Transactions on Cyber-Physical Systems</i> , 8(4), Article 43, 2024.	
	[J4]	N. K. Boran, S. Rathore, M. Udeshi , and V. Singh. "Fine-Grained Scheduling in Heterogeneous-ISA Architectures." <i>IEEE Computer Architecture Letters</i> , 20(1):9–12, 2021.	
CONFERENCE PUBLICATIONS	[C1]	M. Udeshi , V. S. C. Putrevu, P. Krishnamurthy, R. Karri, and F. Khorrami. "An Orchestrated Sandbox for Side-Channel Analysis of Cyber-Physical Systems Malware." <i>IEEE Embedded Systems Letters</i> , presented at <i>Embedded Software Conference (EMSOFT)</i> , 2026.	
	[C2]	M. Udeshi , M. Raz, P. Krishnamurthy, R. Karri, and F. Khorrami. "Sandbox-Enabled Digital Twin for Cyber-Physical Systems." <i>IEEE International Symposium on On-Line Testing and Robust System Design (IOLTS)</i> , 2026.	
	[C3]	M. Shao, N. Rani, K. Milner, H. Xi, M. Udeshi , S. Aggarwal, V. S. C. Putrevu, S. K. Shukla, P. Krishnamurthy, F. Khorrami, R. Karri, and M. Shafique. "Towards Effective Offensive Security LLM Agents: Hyperparameter Tuning, LLM as a Judge, and a Lightweight CTF Benchmark." <i>AAAI Conference on Artificial Intelligence</i> , 40(35):29660–29668, 2026.	
	[C4]	T. Abramovich, M. Udeshi [†] , M. Shao [†] , K. Lieret, H. Xi, K. Milner, J. Yang, C. Jimenez, F. Khorrami, P. Krishnamurthy, B. Dolan-Gavitt, M. Shafique, K. Narasimhan, R. Karri, and O. Press. "EnIGMA: Interactive Tools Substantially Assist LM Agents in Finding Security Vulnerabilities." <i>International Conference on Machine Learning (ICML)</i> , 2025.	
	[C5]	M. Shao [†] , S. Jancheska [†] , M. Udeshi [†] , B. Dolan-Gavitt [†] , K. Milner, B. Chen, M. Yin, S. Garg, P. Krishnamurthy, F. Khorrami, R. Karri, and M. Shafique. "NYU CTF Bench: A Scalable Open-Source Benchmark Dataset for Evaluating LLMs in Offensive Security." <i>Conference on Neural Information Processing Systems (NeurIPS)</i> , 2024.	
	[C6]	N. K. Boran, P. Diwan, M. Udeshi , S. Rathore, and V. Singh. "HIDC: Heterogeneous-ISA Dynamic Core." <i>IEEE International Conference on High Performance Computing and Communications (HPCC)</i> , pp. 284–291, 2024.	
HONORS AND AWARDS	DAC Young Fellowship Design Automation Conference, 2023 Recognition of Outstanding Contributions (RoCStar) Qualcomm, 2021 For the AI100 compiler and KCF projects. Gold Medal, Indian National Physics Olympiad HBCSE, 2014 Awarded to the top 35 students nationwide. Tech Special Mention, among 500+ hostel residents IIT Bombay, 2016		
RESEARCH PROJECTS	Cyber-Physical Systems Security Mar 2025–present <i>Ph.D. research, Control/Robotics Research Lab, NYU</i> Developed a QEMU sandbox for malware dynamic analysis with tailored environment orchestration. Designed an LLM binary diffing framework for software supply chain security, detecting malicious code injection in software updates. Proposed the <i>Ransomware 3.0</i> threat model and prototyped the AI-powered ransomware <i>PromptLock</i>.		

LLM Agents for Cybersecurity

Mar 2024–present

Ph.D. research, Center for Cybersecurity, NYU

Curated NYU CTF Bench with 200 CTF challenges for evaluating autonomous LLM cybersecurity agents. Developed EnIGMA, an LLM agent with interactive debugging tools for vulnerability detection and exploitation. Developed D-CIPHER, an LLM multi-agent framework with collaboration between planner and executor agents.

Reverse Engineering of Cyber-Physical Systems

Aug 2022–present

Ph.D. research, Control/Robotics Research Lab, NYU

Designed REMaQE, a symbolic execution framework for reverse engineering math equations from CPS binaries; achieves 100% accuracy on C and Simulink binaries. Designed REMEND, a Transformer-based neural decompilation framework for reverse engineering equations; achieves 89.8–92.4% accuracy on C and Fortran binaries. Leveraged `angr` for symbolic execution and capstone for binary analysis.

Hardware Security (Masters Thesis)

Aug 2018–June 2019

CADSL, IIT Bombay. Advisor: Prof. Virendra Singh

Designed cache side-channel attacks exploiting prefetchers and the re-order buffer. Built an LLVM-IR framework for x86-to-ARM program migration in the heterogeneous-ISA core HIDC.

TEACHING AND MENTORING	2020–2021	Mentor, hardware security project, Qualcomm Innovation Fellowship.	Qualcomm
	Spring 2019	Teaching Assistant, VLSI Design Lab (Prof. Sachin Patkar).	IIT Bombay
	Fall 2018	Teaching Assistant, Microprocessors (Prof. Virendra Singh).	IIT Bombay
	Summer 2016	Teaching Assistant, CS101 Computer Programming (Prof. D. B. Phatak).	IIT Bombay
	2016	Mentor for 5 participants, Kharagpur Winter of Code.	
SERVICE AND LEADERSHIP	2024–2026	Organizer, LLM Attack Challenge, Cybersecurity Awareness Week (CSAW)	NYU
	2016–2017	Manager, Electronics Club	IIT Bombay
	2015	Reviewer, 46th International Physics Olympiad	Mumbai, India
	2014	Volunteer tutor, Supplemental Learning Program	Vidya NGO Mumbai

Peer Review

AAAI 2026, AAAI 2027, ICLR 2026, NeurIPS 2026, CPS-IoTSec 2025, *ACM Transactions on Intelligent Systems and Technology*, *The Computer Journal*, *The Journal of Supercomputing*, *Journal of Computer Virology and Hacking Techniques*.

SKILLS	Coursework	Hardware Security & Trust, Deep Learning, Advanced Computer Architecture, Advanced Hardware Design, Computer Vision, VLSI Design
	Programming	Python, Embedded C/C++, C, C++, Shell/Bash, Assembly, Verilog, Javascript, HTML, Java
	Techniques	Symbolic execution, static/dynamic binary analysis, neural decompilation, binary diffing, malware sandboxing, side-channel analysis
	Frameworks	<code>angr</code> , Capstone, Ghidra, Binary Ninja, QEMU, ReAct LLM agents, PyTorch, LLVM, PyTorch Glow, gem5, MATLAB/Simulink, Vivado HLS